

AI-Augmented Architectural Design Consulting Firm

Company Structure & Multi-Agent System Architecture — Master Plan

1. Executive Summary

This firm delivers architectural/engineering design services where a **multi-agent AI system** (built on Microsoft Agent Framework) performs the first-draft coordination work across disciplines — architecture, structural, HVAC, electrical, interior, and site survey — and **licensed human professionals** review, correct, and stamp the output before it becomes a deliverable. The business is positioned as *"senior-partner-speed, junior-team-cost"*: clients get multi-discipline coordination in days instead of weeks, with the same professional accountability as a traditional firm.

The company is built in two interlocking tracks that must scale together:

- **Track A — People & Entity:** the licensed professionals, drafters, and business functions required to legally practice and take liability.
 - **Track B — Agents & Platform:** the Supervisor/worker agent system, Revit execution layer, and the engineering team that maintains it.
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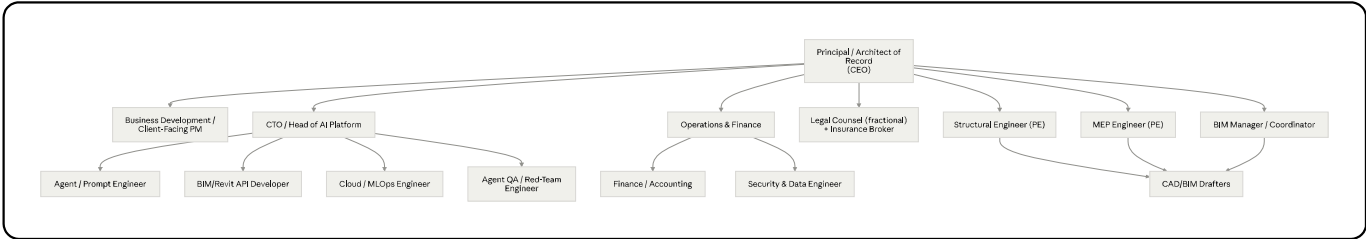
2. Legal & Entity Structure

Decision	Recommendation	Why
Entity type	Professional Corporation (PC) or Professional LLC (jurisdiction-dependent), with a separate Technology LLC (see below)	Most jurisdictions require architecture/engineering firms to be owned/controlled by licensed professionals. Non-licensed technical co-founders cannot hold controlling equity in the PC directly.
Structure	Two entities: (1) the licensed Design PC — holds the stamps, client contracts, liability, insurance; (2) a Technology LLC — owns the agent platform, IP, and Revit/MAF codebase, and licenses it to the PC (and later to other firms)	Separates "practice of architecture/engineering" liability from "software IP" — lets you eventually license the agent platform to other firms without those firms needing your specific licenses, and protects the tech IP if the PC is ever restructured.
Ownership	Licensed Architect/Engineer(s) hold controlling equity in the PC (per local law); technical co-founders hold equity in the Technology LLC, with the LLC and PC cross-linked by services/licensing agreement	Keeps you compliant while still fairly rewarding the technical build.
Principal of Record	One licensed architect (or engineer, depending on primary service line) is legally responsible for all stamped output	Required by law in essentially every jurisdiction — cannot be automated away.

Action item before anything else: confirm with a local AEC-specialized attorney which entity structure is valid in your jurisdiction — this varies significantly by state/country and determines everything downstream (hiring, contracts, insurance).

3. Organizational Structure

3.1 Org chart



3.2 Role summary

Function	Roles	Headcount stage
Licensed practice	Principal/Architect of Record, Structural PE, MEP PE, (Civil Engineer, Fire/Life-Safety Consultant as needed)	Phase 0-1 (must exist before taking any paid client work)
Production	BIM Manager, CAD/BIM Drafters, Quantity Surveyor/Estimator, Specification Writer, Code/Permit Expediter	Phase 1-2
AI Platform	CTO/Head of AI, Agent/Prompt Engineer, BIM-Revit API Developer, Cloud/MLOps Engineer, Agent QA/Red-Team Engineer, Security/Data Engineer	Phase 0-2, scales through Phase 4
Business	Business Development/Client PM, Finance/Accounting, Legal Counsel (fractional), Insurance Broker	Phase 0, ongoing

4. Hiring Sequence (tied to build phases)

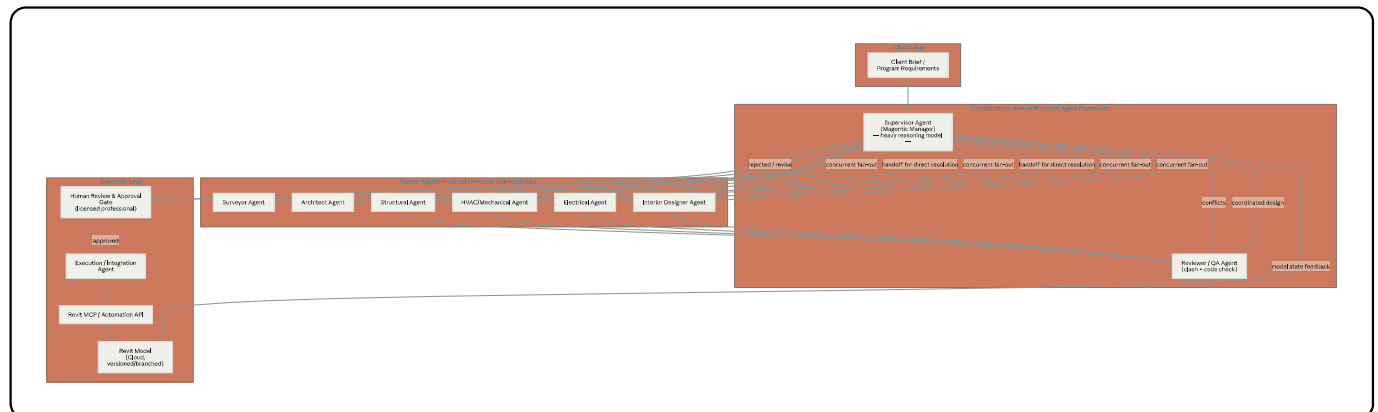
Phase	Company milestone	Hires to make
0 — Discovery & Formation	Form PC + Tech LLC, secure E&O insurance, define pilot project type	Principal/Architect of Record, fractional legal counsel, insurance broker, CTO/technical co-founder
1 — Single-agent PoC	Prove Revit execution path with human review	BIM Manager, BIM/Revit API Developer, 1 CAD/BIM Drafter
2 — Supervisor + 2 workers	First real (small) client pilot	Structural PE or MEP PE (whichever pilot needs), Agent/Prompt Engineer, Business Dev/Client PM
3 — Full discipline team	Multi-discipline coordinated projects	Remaining PE(s), Quantity Surveyor, Specification Writer, Agent QA/Red-Team Engineer, Security/Data Engineer
4 — Productize	Repeatable service lines, possible platform licensing to other firms	Finance/Accounting, Code/Permit Expediter, additional drafters/PMs scaled to project volume

5. Multi-Agent System Architecture

5.1 Design principles

1. **Supervisor decomposes and decides; workers execute narrowly.** No worker agent should have a broad, generalist scope — each is bound to one discipline's tools and context.
2. **Only one path writes to Revit.** Workers never touch the live model directly; a single Execution layer, gated by human approval, does.
3. **Every agent decision is logged against the model element it affects.** Traceability is a legal requirement here, not a nice-to-have.
4. **Mixed model sizing.** Supervisor/manager runs on a stronger reasoning model; narrow, repeatable worker tasks run on faster/cheaper models.
5. **Human sign-off is a hard gate, not a suggestion,** before anything construction-relevant is written or issued.

5.2 Architecture diagram



5.3 Agent roster and tool scope

Agent	Orchestration role	Bound tools/data	Escalates to human when
Supervisor (Magentic manager)	Builds plan, assigns tasks, judges progress, decides completion	Task ledger, agent capability registry, project brief	Ambiguous or conflicting client requirements
Surveyor	Worker	Site/GIS data, survey scans	Site data incomplete or contradictory
Architect	Worker	Code/zoning DB, space program, massing tools	Code conflicts, egress issues
Structural	Worker	Load tables, structural code refs	Any load-bearing decision (always human PE sign-off)
HVAC/Mechanical	Worker	Load calc tools, equipment specs	Any load-bearing/life-safety equipment sizing
Electrical	Worker	Panel/circuit standards, code refs	Life-safety systems (fire alarm, emergency power)
Interior Designer	Worker	Finish/material libraries, furniture catalogs	Budget-relevant material substitutions
Reviewer/QA	Cross-cutting	All worker outputs, clash-detection rules	Unresolvable cross-discipline conflict
Execution/Integration	Gated executor	Revit MCP tool set only	Any write outside pre-approved scope

5.4 Orchestration pattern mapping (Microsoft Agent Framework)

MAF builder	Used for
MagenticBuilder	Supervisor's overall dynamic planning and delegation — the backbone pattern
ConcurrentBuilder	Parallel drafting once Architect's base layout is fixed (HVAC/Electrical/Interior run simultaneously)
HandoffBuilder	Direct specialist-to-specialist resolution (e.g., Electrical ↔ Architect on ceiling void) without round-tripping through the Supervisor
Tool-approval / request-info	The human-in-the-loop gate before any Revit write

5.5 Data & execution flow (step by step)

1. Client brief → Supervisor.
2. Supervisor tasks Surveyor, then Architect, sequentially (each depends on the last).
3. Supervisor fans out to Structural, HVAC, Electrical, Interior concurrently once the base layout exists.
4. Reviewer/QA agent checks all outputs for clashes and code conflicts; unresolved conflicts go back to Supervisor, who may invoke Handoff between two specific agents.
5. Supervisor synthesizes a single coordinated instruction set.
6. **Human review gate** — licensed professional reviews before execution.
7. Execution/Integration agent translates approved instructions into Revit MCP/Automation API calls against a branched/cloud model (never the live production file).
8. Updated model state feeds back to Supervisor for the next revision cycle or client review.

5.6 Technology stack summary

Layer	Technology
Agent orchestration	Microsoft Agent Framework (Python or .NET), <u>agent-framework-orchestrations</u>
Hosting/models	Azure AI Foundry, mixed model sizing (reasoning model for Supervisor, lighter models for workers)
Revit integration	Autodesk Automation API for Revit + MCP tools bundle (preferred); community Revit MCP servers for prototyping only
Human-in-the-loop	MAF tool-approval / request-info pause mechanism
Logging/traceability	Per-agent, per-tool-call audit log tied to specific model elements
Data/security	Isolated per-client environments, access control on model data, retention policy

6. Governance, Risk & Liability

- **Chain of custody:** every stamped deliverable maps to one named licensed human accountable for the corresponding agent's output (Structural PE ↔ Structural Agent, etc.).
- **Insurance:** Professional Liability/E&O coverage explicitly structured to cover AI-assisted design work — confirm with a broker who has handled this category before binding.
- **Disclosure policy:** define now whether/how AI involvement is disclosed to clients and regulators; treat as a live regulatory question, not a settled one.
- **Model integrity:** agent-driven changes always target branched/cloud models; production files are only updated after human approval.
- **QA/red-teaming:** a human QA function whose job is deliberately breaking the agent system (malformed briefs, conflicting instructions, edge-case geometry) before a client does.

7. Service & Engagement Model

- **Positioning:** multi-discipline coordination delivered in days, at junior-team cost, with senior-firm accountability.
- **Engagement structure:** client brief intake (structured, not freeform) → AI-coordinated first draft → human review/revision cycles → stamped deliverable.
- **Pricing considerations:** blend of project-based fees (traditional AEC norm) with usage-based cost accounting for AI compute — track this separately from billing to

avoid margin surprises as agent usage scales with project complexity.

- **Pilot strategy:** start with a narrow, low-risk project type (e.g., small commercial tenant fit-out) to prove the Phase 1–2 pipeline before taking on complex multi-building or life-safety-heavy work.

8. Build & Company Roadmap (combined)

Phase	Technical milestone	Company milestone
0 — Discovery	Define brief format, Revit ops needed	Form PC + Tech LLC, hire Principal + CTO, secure E&O insurance
1 — Single-agent PoC	Architect agent writes to test Revit model via MCP, human review	Hire BIM Manager, Revit API Developer, 1 drafter
2 — Supervisor + 2 workers	Magentic manager + 2 disciplines, human approval gate	First pilot client, hire relevant PE + Agent Engineer + Business Dev
3 — Full team	All disciplines, Reviewer/QA agent, clash detection	Hire remaining PEs, QA/red-team engineer, security engineer
4 — Productize	Logging/traceability hardened, pricing model live	Finance hire, scale drafters/PMs to volume, evaluate licensing the platform to other firms

Next steps: confirm entity structure with local counsel (Section 2), then begin Phase 0 hiring and pilot project selection in parallel with the single-agent Revit PoC.